Entropy Aware Training for Fast and Accurate Distributed GNN

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Introduction



Social Network
(Facebook Page-Page Network)



Citation Network
(Cora, PubMed, OGB-Arxiv)

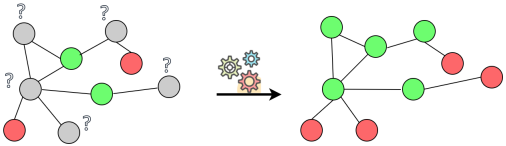


Network Graph



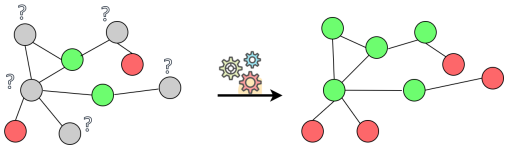
Communication Network

Graph Neural Networks

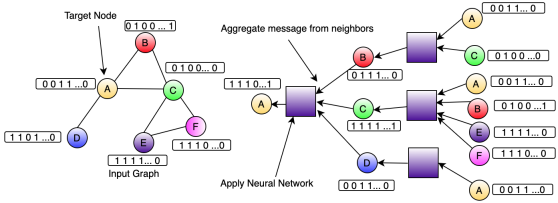


Node Level Task

Graph Neural Networks

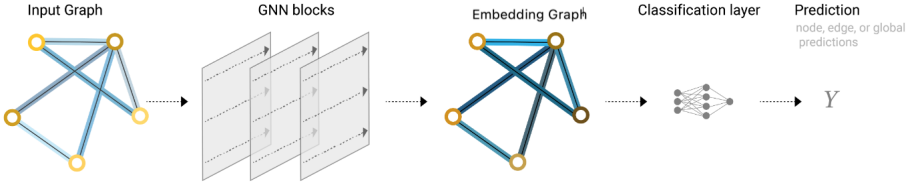


Node Level Task



Message Passing

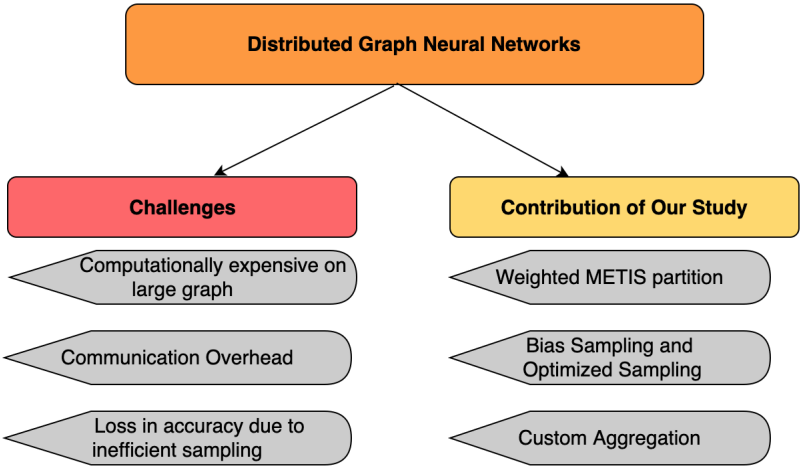
Graph Neural Networks



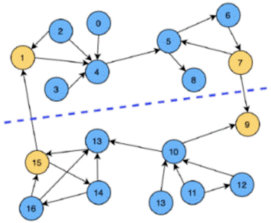
Prediction task with GNN models

Image Source: https://distill.pub/2021/gnn-intro/

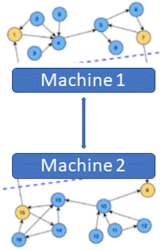
Motivation of Distributed GNNs



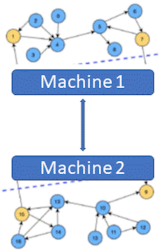
Scaling Graph Neural Networks



Input Graph is **partitioned** and sent to multiple machines



Embeddings of **remote** neighbors are **communicated** to complete **Aggregation Step**



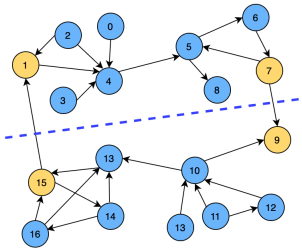
Each Machine trains its own GNN and **predicts** test nodes

Edge weights to preserve low degree neighborhoods (communities)

Biased and Optimized Sampling to reduce remote neighbors

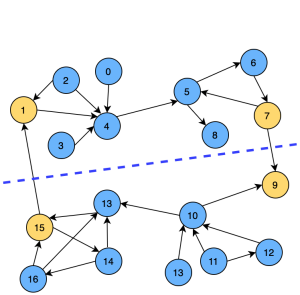
Custom aggregation emphasizes information from remote neighbors (Weak Ties)

Weighted Graph Partitioning (Prefer low degree nodes)

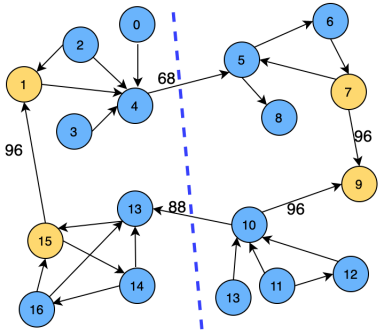


METIS Partition

Weighted Graph Partitioning (Prefer low degree nodes)



METIS Partition

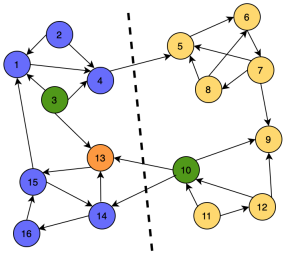


Weighted METIS Partition

$$P_{uv} = 1 - \left(1 - \frac{1}{|N(v)|}\right)^{k} \quad (1)$$

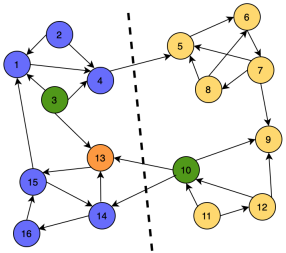
$$W_{uv} = P_{uv} * 100 \quad (2)$$

Bias Neighborhood Sampling (Prefer local nodes)

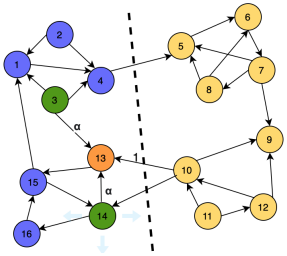


Random Neighbors sampling
(For 1-hop sampling with k=2
for target node=13)

Bias Neighborhood Sampling (Prefer local nodes)

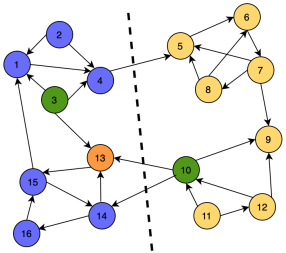


Random Neighbors sampling (For 1-hop sampling with k=2 for target node=13)

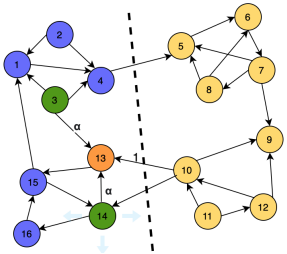


Bias Neighbors sampling (For 1-hop sampling with k=2 for target node=13.)

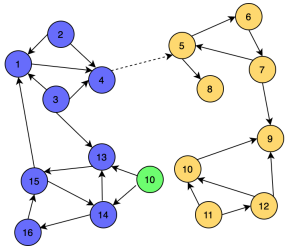
Bias Neighborhood Sampling (Prefer local nodes)



Random Neighbors sampling (For 1-hop sampling with k=2 for target node=13)



Bias Neighbors sampling (For 1-hop sampling with k=2 for target node=13.)



Optimized Neighbors sampling

Custom Aggregation Function

- Usual Aggregation function.

$$h^i_{N(v)} = AGG_i(h^{i-1}_u, \forall u \in N(v)) \tag{3}$$

- Bias sampling may suffer from information loss.
- Node receives more message from local nodes and less from remote nodes.
- We assign higher weight (by the factor α) to remote nodes messages[1].

$$h^i_{N(v)} = AGG_i(\boldsymbol{\alpha h^{i-1}_r}, h^{i-1}_l; \forall r \in R(v), \forall l \in L(v)) \tag{4}$$

$$h^i_v = \sigma(h^i_{N(v)}, h^{i-1}_v) \tag{5}$$

$L(v)$: Neighbors of v from local partition, $R(v)$: Neighbors of v from remote partition

[1] The Strength of Weak Ties, Mark S. Granovetter

Experimental Setup

- Benchmark:-
  - Social Network: Facebook Large Page-Page Network
  - Citation Network: Cora, OGB, and PubMed
- Baseline:-
  - DistDGL: Distributed Graph Neural Network Training for Billion-Scale Graphs
- Experiment Conducted on AWS:-
  - Deep Learning AMI (Ubuntu 18.04) Version 59.0, 64-bit(x86)), Intel(R) Xeon(R) CPU Es-2686 v4, 8 processing core with 2.30GHz frequency
  - 32GB RAM, 200GiB EBS volume
- Framework:-
  - Deep Graph Library (0.7 version)
- GNN Model and Hyper Parameters:-
  - GraphSage
  - Alpha=1,5,10, k=25, n_layer=2, and lr=0.003

DistDGL: https://arxiv.org/abs/2010.05337

Dataset

Data Set	Nodes	Edges	Features	Labels	Train/Test/Val
Cora	2708	5429	1433	7	140/1000/500
PubMed	19717	44338	500	3	60/1000/500
Facebook	22470	171002	NA	2	NA
GIT-ML	37700	289003	NA	2	NA
OGB-Arxiv	132534	1166243	128	40	48603/390941/29779

Dataset Statistics





Effect of Bias (α) in Biased Sampling

Dataset	Alpha (α)	M+RS	WM+BS
Cora	1	0.14	0.13
	5	0.14	0.14
	10	0.14	0.08
PubMed	1	0.12	0.12
	5	0.12	0.08
	10	0.12	0.06
Facebook	1	0.83	0.83
	5	0.83	0.77
	10	0.83	0.71
GIT-ML	1	0.75	0.75
	5	0.75	0.68
	10	0.75	0.62
OGB-Arxiv	1	0.43	0.43
	5	0.43	0.41
	10	0.43	0.39

Ratio of Proxy Vertices to total graph nodes with different values of parameter α.
M: METIS partitioning, **RS**: Random Sampling, **WM**: Weighted METIS partition,
BS: Bias Sampling, Number of Partition=4



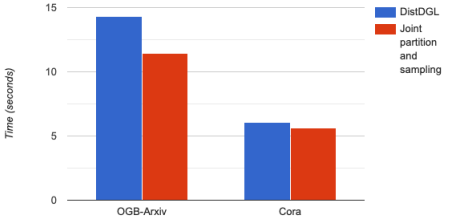


Effectiveness of Joint Partitioning and Sampling Algorithms

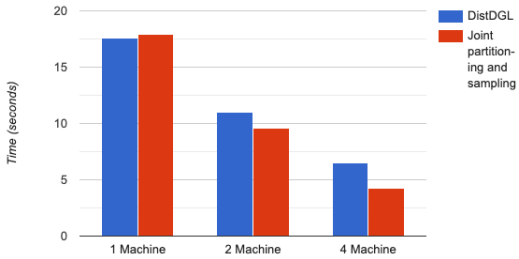
Dataset	#Partitions	M + RS	WM + BS	WM + OS
Cora	2	0.1	0.05	0.06
	4	0.13	0.08	0.06
	8	0.21	0.12	0.04
	16	0.25	0.15	0.03
PubMed	2	0.08	0.04	0.04
	4	0.12	0.06	0.04
	8	0.21	0.11	0.04
	16	0.26	0.15	0.04
OGB-Arxiv	2	0.35	0.31	0.26
	4	0.43	0.38	0.29
	8	0.51	0.46	0.29
	16	0.58	0.52	0.31

Effectiveness of Joint Partitioning and Sampling Algorithms
α=10. **M** : METIS, **RS** : Random Sampling, **WM** : Weighted METIS,
BS : Bias Sampling, **OS** : Optimized Bias Sampling

Overall Runtime and Scaling



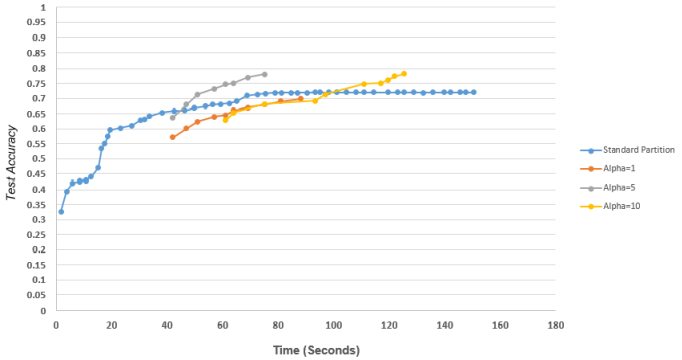
The overall run time per epoch (batch size is 2k)



GNN Scaling Behavior on OGB-Arxiv Dataset

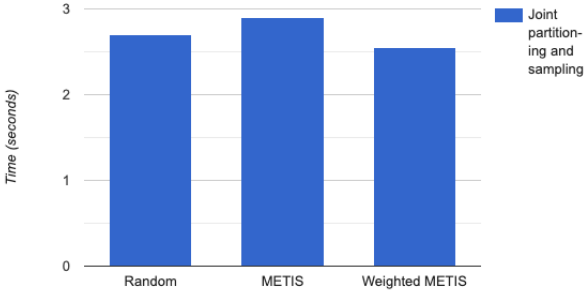
Distributed GNN Test Accuracy

- Our approach achieves better accuracy after 42 sec at alpha=5



Test accuracy of distributed GraphSage on 4 machines with OGB-Arxiv

Comparing Different Partitioning Policies



Comparing partition policies using 4 machines with OGB-Arxiv

Conclusion

1. Joint Partitioning and Sampling algorithm for distributed GNN
2. Reduction communication overhead by 53%
3. Less partitioning time to partition graph
4. Improved accuracy
5. Speedup of upto 1.5x on OGB-Arxiv





Questions **?**

Thank You



